

Szymon Galutowski

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EDUCATION

University of Warwick

Computer Science Integrated Masters, Predicted 1st Class Hons

- Graduation expected in 2029 but able to switch to Bachelors and graduate in 2028

Coventry, England

Oct. 2025 – 2029

Woodhouse College

*Maths, Further Maths, Computer Science, Physics (all A-A*s)*

London, England

Sep. 2023 – May 2025

EXPERIENCE

Research Intern

WMG, University of Warwick

- Engineered synthetic datasets using **GANs** to train and evaluate **Federated Learning** models.
- Analysed model weights, confusion matrices, and activation heatmaps across **MNIST** and **SVHN** datasets.
- Identified variance in **L2 norms** between synthetic and real-world data to improve model security against adversarial attacks.

Jul. 2026 – Sep. 2026

Coventry, England

Data Annotator

Data Annotation

- Constructed adversarial JSON test cases to probe AI models for edge-case failures, evaluating responses to produce human-feedback training data.

Jun. 2025 – Sep. 2025

Westminster, London

Work Shadowing Intern

Wise & Surfboard

- Developed a Python script utilising **Pandas** to analyse and extract insights from financial trends.
- Researched and presented on money transfer **security flaws**, proposing actionable architectural mitigations.

Jul. 2024

Westminster, London

PROJECTS

Gesture Recognition Calculator | *Python, JavaScript, MediaPipe, OpenCV*

- Developed a touchless calculator interpreting physical hand gestures in real-time using **computer vision**.
- Curated and annotated a **custom dataset** of 1,000+ images to train **classification models** for digits and mathematical symbols.
- Ported the Python machine learning inference logic into **client-side JavaScript** for zero-backend browser deployment.

Szymon's Movie Database | *Java*

- Built a high-performance **in-memory database engine** from scratch, with custom indexing delivering efficient exact-match and range queries over a 1,000-film dataset.
- Implemented custom data structures without standard libraries, including a dynamically resizing **hash map** and self-balancing **AVL trees**.
- Optimised range queries to **$O(\log n + k)$** time complexity and developed a generic **Quicksort** algorithm for reliable $O(n \log n)$ sorting.

λ GPT: Functional Chatbot | *Haskell*

- Built a terminal-based conversational assistant structuring the REPL within a **StateT monad transformer** stack to fuse state manipulation with I/O.
- Implemented persistent key-value memory with disk rehydration, supporting associative recall and targeted forgetting.
- Integrated the Free Dictionary **REST API** via **http-conduit**, deriving **FromJSON** (Aeson) instances to parse multi-level responses.

Further projects at szymonator.github.io/portfolio

TECHNICAL SKILLS

Languages: Python, Java, Haskell, JavaScript, SQL (Postgres), HTML/CSS, TypeScript

Frameworks: React, Node.js, Flask

Developer Tools: Git, VS Code, Antigravity

Libraries: Selenium, Pandas, NumPy, Matplotlib, OpenCV, MediaPipe